

COURSE CODE : DSA0601 Data handling and Visulazation

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Set 10: Survey Responses Analysis

Data Visualization using R

Aim

To analyse survey responses using a grouped bar chart, a stacked bar chart, a table and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the survey data as a data frame.
3. Question 1: draw the grouped bar chart of Question 1 answers with `geom_bar(position = "dodge")`.
4. Question 2: draw the stacked bar chart of all three questions with `geom_bar()`.
5. Question 3: make the table with `tableGrob()`.
6. Question 4: join the two charts and the table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
survey <- data.frame(Survey_ID = 1:3, Question1 = c("A","B","C"),
                     Question2 = c("B","A","A"), Question3 = c("C","D","B"))
print(survey)
long <- stack(survey[, 2:4]); names(long) <- c("Answer", "Question")
print(table(long$Question, long$Answer))

# 1. Grouped bar chart - Question 1
p1 <- ggplot(survey, aes(Question1, fill = factor(Survey_ID))) +
  geom_bar(position = "dodge") +
  labs(title = "Answers of Question 1", x = "Answer", y = "Count", fill = "Survey ID")

# 2. Stacked bar chart - all questions
p2 <- ggplot(long, aes(Question, fill = Answer)) + geom_bar() +
  labs(title = "Responses of All Questions", x = "Question", y = "Count")

# 3. Table
tb <- tableGrob(survey, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Survey Dashboard")

ggsave("set10_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set10_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set10_q3.png", arrangeGrob(tb, top = "Survey Response Table"), width = 5, height = 2, bg = "white")
ggsave("set10_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

Survey_ID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

	A	B	C	D
Question1	1	1	1	0
Question2	2	1	0	0
Question3	0	1	1	1

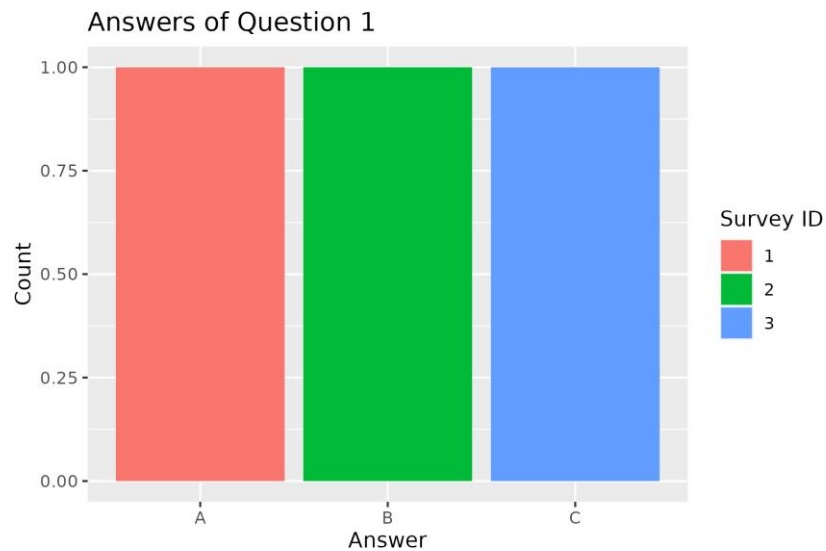


Figure 10.1: Q1 - Grouped bar chart (Question 1)

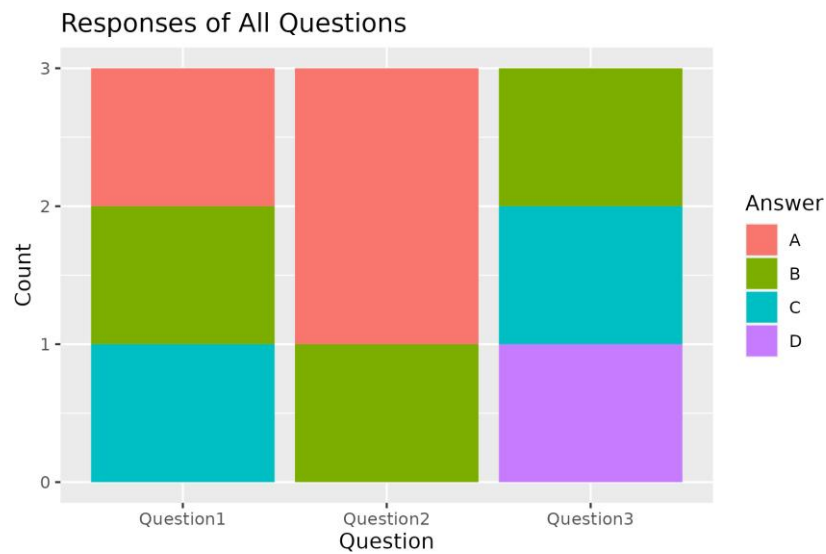


Figure 10.2: Q2 - Stacked bar chart (all questions)

Survey Response Table

Survey_ID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 10.3: Q3 - Survey response table

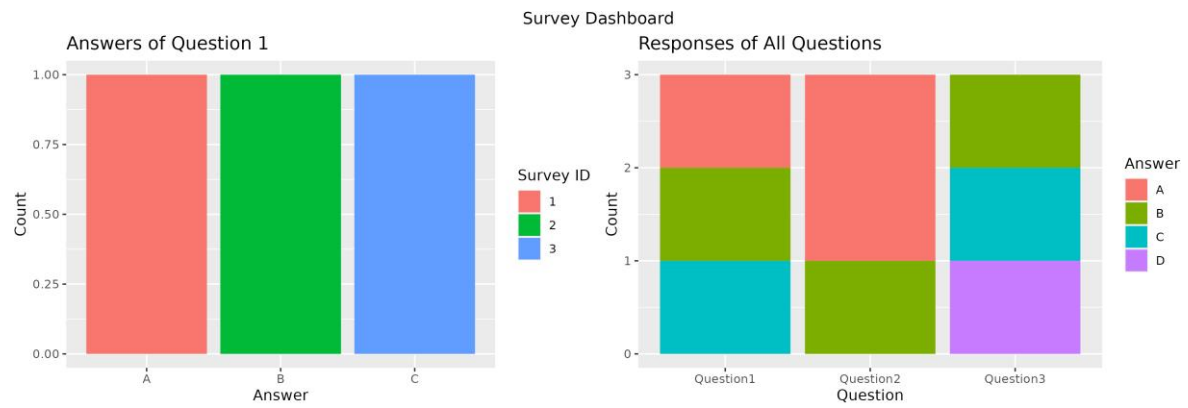


Figure 10.4: Q4 - Dashboard

Result

- In Question 1, answers A, B and C were chosen once each.
- Question 2 has A twice and B once; Question 3 has B, C and D once each.
- All four parts were completed successfully in R.

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